

17/01/2020

# From Brain Signal to Language [Paul Smolensky]

Paul Smolensky

What we are at:

→ New kind of existence to co-exist with

→ "Final Revision" story to Claude

↳ NN, "neurons connected with each other" "Connectionism"

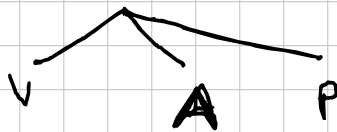
SYMBOLS: INPUT & OUTPUT → symbols, where  
↳ recurrent activation vector

↳ How does it look from the p.o.v of COGNITIVE science?

COGNITION  $\approx$  COMPUTATION

A Pendulum...

Discrete symbolic structures (1955)  
(manipulated by rewrite rules)



Numerical Vectors (1985)

PDP generation

Discrete symbol structures 1985  
A = assigned probabilities

Numerical Vectors

Deep Learning generation (2005)

What kind of computational system is cognition?

↓  
symbolic computation to understand models

↘ Neural Comp, LLM

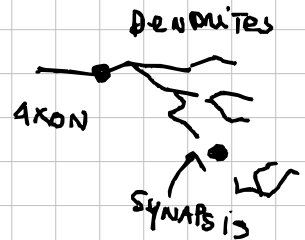
What is The secret of human intelligence?

PROGRAM: MIND

- REASONING
- MATH
- NLP

HARDWARE: BRAIN

- General properties
- Detailed properties



Discovering The implicit symbolic structure of LLM

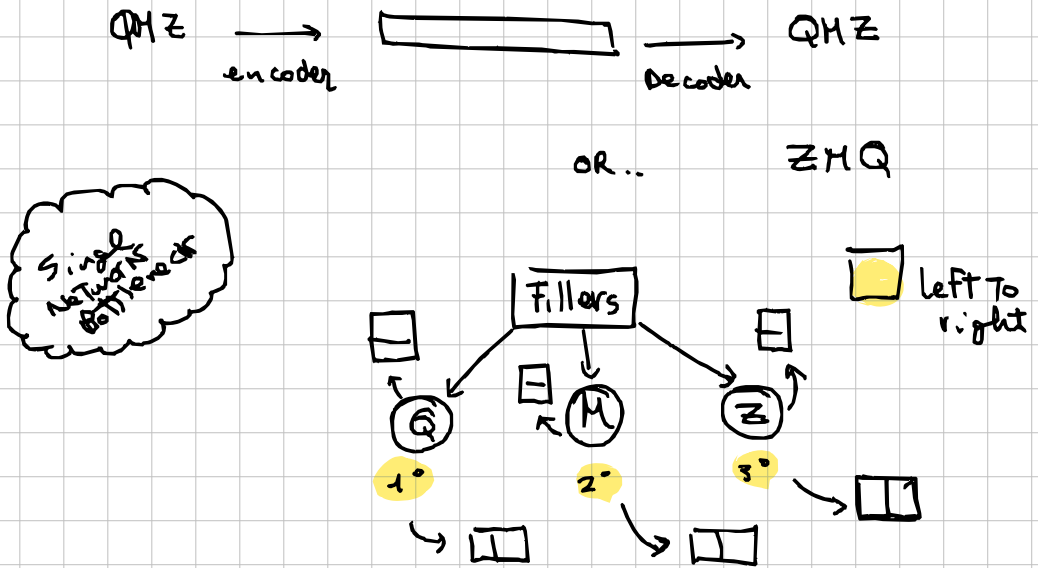
(Tom McCoy)

Dog chase cats  
↑  
Cats chase Dog

1 0 1 0 0 1

1 0 1 0 0 1

HP: They encode Tensor Product Representation



Tensor products of  $F$  and  $\text{[Box]}$   $\rightarrow$  LINEAR MAP

NN are implicit TPRrepresenters

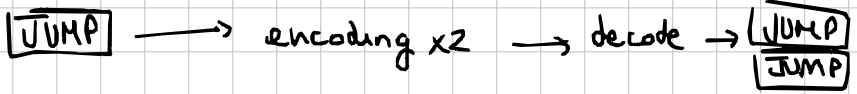
We Feed The TPRs To a decoder To check

MLP  
RNN  
Transformer

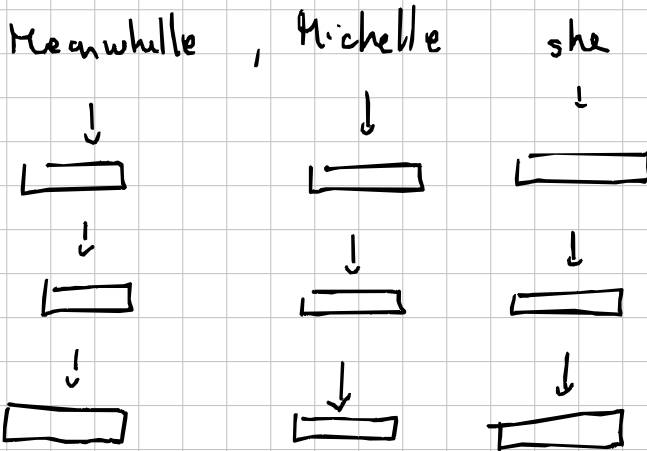
} Turns out BIDIRECTIONAL WORK 100%  $\forall$  architecture

# UNcovering TPR in Embed Representation, (Tom McCoy)

Jump twice

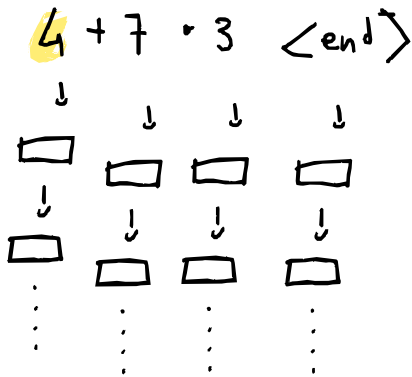


## Prompting Claude



D. TPR appear in LLM as well?

- Arithmetic
- Logic
- Linguistic Transformations
- ...



Sequential  
SYNTACTIC

We construct  
hypothesis on  
STRUCTURE

(Position, Pos ...)

25

TPR in GA-OSS

Summe performances  
on these tasks

Can we do causal intervention? ✓

The clever doctor helps the lawyer

- clever: subject-adjective + clever: object-adjective

TPR explains other interpretability methods ✓

Also TPR connect

NN theory  
(BRAIN)

→

SYMBOLIC  
RULE THEORY  
(MIND)

18/08/26

TPR enable complex processing of discrete language structures

1) How To explain human cognition?

Cognition is computation

How?

- Explicitly encodes and applies symbolic rules  
Symbolic rules can not cover everything

↳ Symbolic residue

Ex. Next-word prediction vs Subj=verb dyroemen

Cognitive Architecture

Foundation : Cognition in computation

What is computation?

|                       |   |              |
|-----------------------|---|--------------|
| □ Cognitive Functions | { | Pixels       |
|                       |   | Sound Stream |
|                       | { | word string  |
| HIGHER                |   | Lexical Form |

Computational Architecture : primitive + combinations

## → Pure Symbolic Architecture (PSA)

- Data: strings, binary trees

- Operations

### Primitives

- Concatenate = cons
- First member = ex 0/1

### Combinators

- Composition  $f(x) = g(h(x))$
- Conditional branching

We have Discrete data (is or is not X)

EX. f passive → LF

Few leaders are admired by Donald Trump

## The Connectionist hypothesis

FDP COMPUTATION

- Representation : Distributed activation vectors
- Primitive operations
  - Multiplication
  - Summation
  - Non-linear transfer function

$$a_\alpha = f(\sum_p w_{p\alpha} a_p) = F([W a]_\alpha)$$

- Combinations : parallelism

→ Data and operations of connectionism do not promise such capability

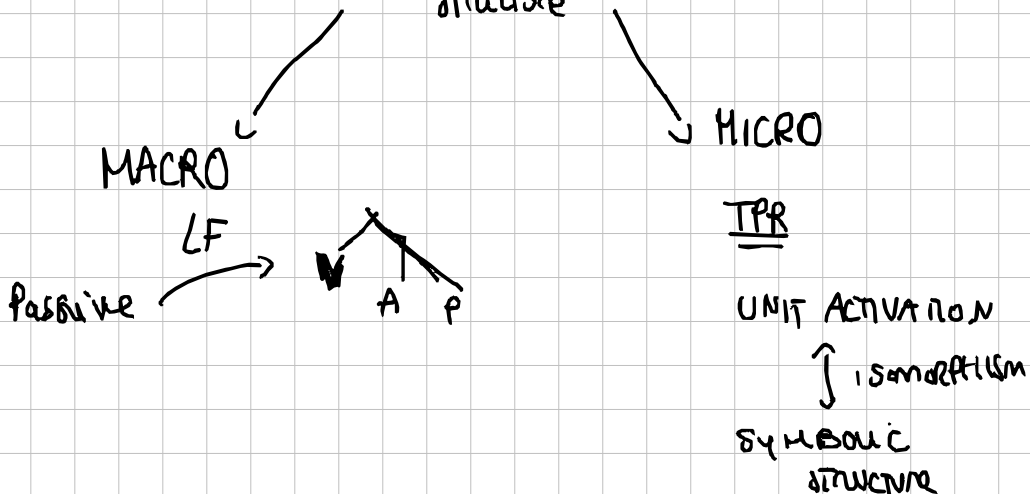
Trying to solve this problem: NECST

## The NECST Hypothesis

- In the core of HD domain, representation and function are well approximated by symbolic
- Connectionist hyp is correct
- Cognitive theory must reduce to PDP computation

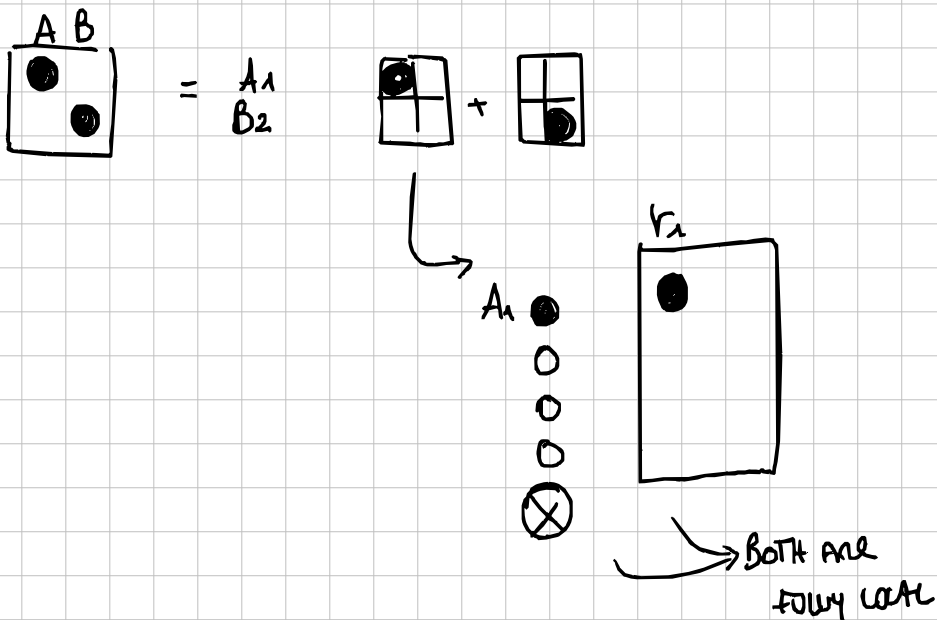
## NECST Isomorphism

NECST Hypothesis: distinguish micro to macro structure

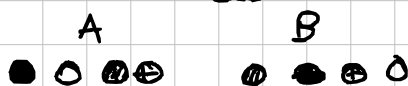




# Fully local representation of letter strings



## Semi-local



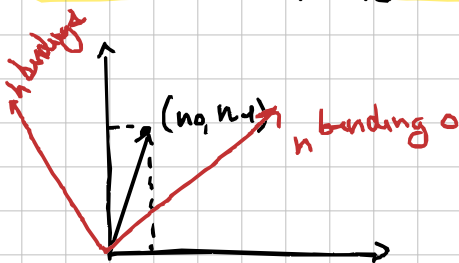
## Fully distributed

$$AB = A \otimes r_1 + B \otimes r_2 = \sum_{x \in F} x \otimes r_x$$

no linear map  $W$

- proper TFR: vectors for all fillers and all roles are linearly independent
- Perfect unbinding by inner product  $\rightarrow$  can get fillers

# From neural to conceptual coordinates

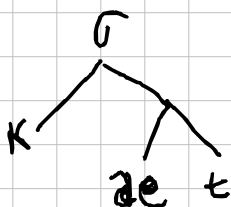


→ Map from activations to neuron to activations of concepts

Tensor product of Atoms

$= \sum_k \text{internal state} \otimes \text{vector for orbital of electron } k$

[TPR exhibit error analogues with King-men: queen-woman]



MACRO STRUCTURE

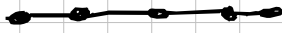
$\sigma/r_3$

$k/r_0$

$ae/r_{01}$

$t/r_{11}$

MICRO



$$\sum_i f_i \otimes r_i$$

Filler  $\otimes$  role = filler/role      bind a filler to a role  
 isomorphism      micro  $\leftrightarrow$  macrostructure

Recursion       $r_0 \otimes r_x = r_{\text{left-child}}$   
 $r_x \otimes r_x = r_{\text{right-child}}$

## TYPE OF ROLES IN TPRS

Define:      Structural Type {role}  
                  Structural token {filler/roles}

| <u>EX.</u> | <u>EXAMPLE</u> | <u>CONTEXTUAL</u>                    | <u>POSITIONAL ROLE</u>                                                           |
|------------|----------------|--------------------------------------|----------------------------------------------------------------------------------|
|            | K runs         | K / runner<br>$K \otimes \text{run}$ | K / agent, run / V<br>$K \otimes \text{agent} + \text{run} \otimes V$            |
|            | J loves K      | $J \otimes \text{loves} \otimes K$   | $J \otimes \text{agent} + \text{love} \otimes V$<br>+ $K \otimes \text{patient}$ |
|            | [BC]           | $B \otimes C$                        | $B \otimes \text{ro} + C \otimes V_1$                                            |
|            | [A[BC]]        | $A \otimes (B \otimes C)$            |                                                                                  |

## Computations over TreePR

Hand Picked

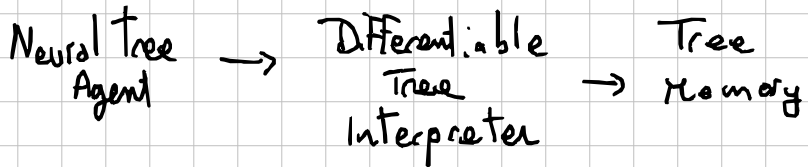
Can we do real symbolic processing with TreePR?

Semantics:  $\beta$  Reduction

Syntax: Tree Adjoining

Learned

## Differentiable Tree Machine



Neural Tree Agent

③ NECS-Theory: discrete compositionality and grammar

General Symbolic Structures

[STRINGS ✓]  
[TREES ✓]

What is a symbolic structure?  
Has a number of ROLES, each role containing a SYMBOL

$\{R_1: S_1, R_2: S_2, \dots\}$

What primitives to embed in the DL model  
Symbolic structures??

TPAs neurally embed the primitives of symbolic computations

- 1) binding
- 2) operation: binding, aggregation, unbinding
- 3) Fillers contain roles
- 4) Fillers can be structures
- 5) Roles can be recursive
- 6) Roles and fillers are mapped independently

→ Compositionality challenge for Connectionism

↳ to develop an effective micro-level formalism for compositionality that resides in vector spaces

|| NNV don't intuitively understand that the world is compositional!

→ encodings that are

- Compositional
- Continuous

"FRIEND" in French

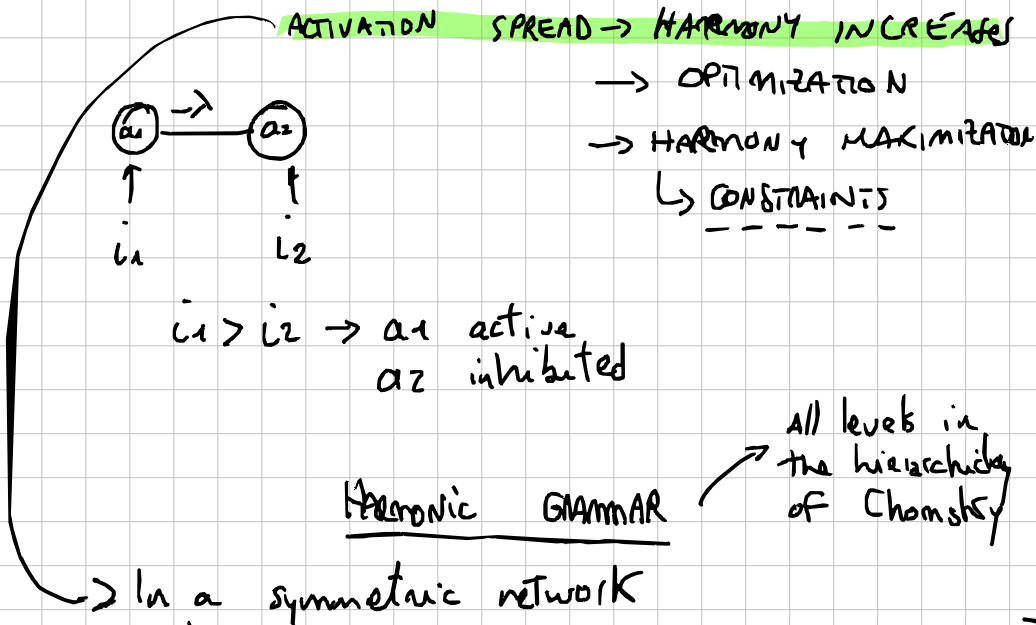
$[0.37 + 0.8n + 0.3t]ami$

Ex. "LOCKABLE"

LOCK [ , - ] & ABLE [ -, ]  
LOCK: L ABLE: R

# ACTIVATIONS - DISCRETE COMPOSITIONALITY and GRAMMAR

$$\frac{da_1}{dt} = -a_1 + i_1 - \lambda a_2$$



$$\frac{dH}{dt} \propto \sum_i \frac{da_i}{dt} [i_i] \geq 0$$

$$\left[ \text{sign} \frac{da_i}{dt} = \text{sign } i_i \right]$$

What does  $H_{\max}$  with TPR?

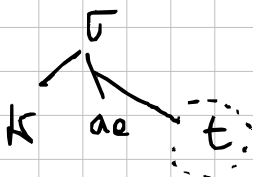
$\rightarrow$  MICRO-LEVEL

$\hookrightarrow$  MACRO-LEVEL

$$H = \sum_{\alpha\beta} a_{\alpha} W_{\alpha\beta} a_{\beta}$$

Constraints

No Coda  $H < 0$   
[Hawai, French, English]



Violation

# Grammar of symbol structures

$\text{wellformedness} \uparrow \text{optimization} \uparrow \text{gradience}$   
Neural Network Computation

(HARMONIC GRAMMAR  $\rightarrow$  OPTIMALITY THEORY)

Discrete Harmonic Grammar : Discrete Input & Output

I want to express <RAINS>

|             | <u>SUBJECT</u> | <u>FULL INTERP</u> |    |
|-------------|----------------|--------------------|----|
| rains       | *              |                    | -3 |
| it rains    |                | *                  | -2 |
| it rains it |                | **                 | -4 |
| rains it    | *              | *                  | -5 |



Dynamics = SGD + quant. straight

$\rightarrow$  CONSTRAINTS (different in my language)



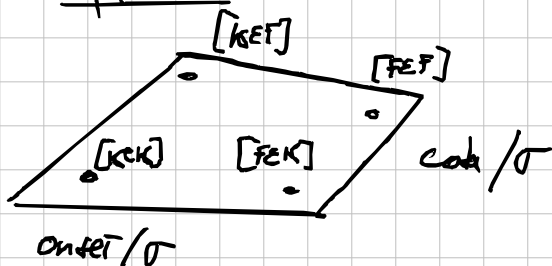
|| In the limit  $t \rightarrow \infty$   
 $T \rightarrow \sigma$   
 || TPR optimize Harmony

[ Incremental sentence processing ]

# NON DISCRETE STRUCTURE OF COGNITION

Gradience (vector space structure) Continuous

Ex. Syllables



Phonology not  
phonetics

$$\begin{matrix} \nearrow & \nwarrow \\ 0.7A & 0.4A \\ +0.2B & 0.9C \end{matrix}$$

$$0.7A \otimes r_{left} + 0.4 \otimes r_{right} \\ + 0.2B \otimes r_{left} + 0.9C \otimes r_{right}$$

3 Repr of a sentence  $\begin{cases} \rightarrow \text{Phonological} \\ \rightarrow \text{Logical} \\ \rightarrow \text{Syntax Structure} \end{cases}$

What did Kay V?

Fix. LF:  $Q_{What} . V(Kay, X_{What})$

↓ optimize  
SS

SS: What Kay V

↓ optimize  
PF



## Gradient: REPRESENTATIONS IN GRAMMAR: LIASON -

IF we give up on DISCRETE Representations  
Language may be better.

- /peti:t + ami/ & peti + ami (syllabic)  
(orthography)

2 COMPETING  
DISCRETE ANALYSIS

Structural blend hypothesis

LLM have  
continuous